

Spectral invariance on the compact core and a nonabelian full subcase of the weighted L^p symmetry conjecture

Candidate partial-result packet

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Status. Candidate substantial partial result, likely valid pending expert review.

1 Source conjecture

Let G be a compactly generated locally compact group of polynomial growth, $1 < p < \infty$, and $1/p + 1/q = 1$. A continuous symmetric weight $\omega: G \rightarrow [1, \infty)$ satisfies (LPAIlg) when, up to harmless positive constants,

$$\omega(xy) \leq \omega(x)\omega(y), \quad \omega^{-q} * \omega^{-q} \leq \omega^{-q}.$$

Then $\mathcal{A} = L^p(G, \omega)$ is a Banach $*$ -algebra for convolution and embeds continuously into $L^1(G)$. For a relatively compact symmetric generating neighbourhood U , put

$$s(n) = \sup_{x \in U^n} \omega(x).$$

Condition (S) is $s(n)^{1/n} \rightarrow 1$; on groups of polynomial growth it is equivalent to the GRS condition [1, 2].

Kuznetsova and Molitor-Braun prove symmetry when G is abelian and, for general G , when ω is polynomial in Pytlik's sense. They then ask whether (LPAIlg) plus (S) suffices in general [1, p. 12].

$$r_{p,\omega}(f) \leq \lim_{n \rightarrow +\infty} \|f^{*n}\|_1^{\frac{1}{n}} = r_1(f).$$

□

We finally get:

Theorem 3.9. *Let (G, ω) satisfy (LPAIlg). Let us assume that ω is polynomial in the sense of Pytlik, i. e. that*

$$\omega(xy) \leq C(\omega(x) + \omega(y)), \quad \forall x, y \in G.$$

Then $L^p(G, \omega)$ is a symmetric Banach $$ -algebra.*

Proof. This follows from Lemma 3.1 of ([Fe-Grö-Lei]) applied to $\mathcal{A} := L^p(G, \omega)$ and $\mathcal{B} := L^1(G)$, and from the result of Losert ([Lo]) about the symmetry of $L^1(G)$. As a matter of fact, these results imply that

$$\sigma_{L^1(G)}(f) = \sigma_{L^p(G, \omega)}(f), \quad \forall f = f^* \in L^p(G, \omega),$$

and, as $L^1(G)$ is symmetric, $\sigma_{L^1(G)}(f) \subset \mathbb{R}$. □

Example: We know that for $D > 0$ sufficiently large, $\omega(x) = (1 + |x|)^D$, where $|x| = \inf\{n \mid x \in U^n\}$, gives rise to an L^p -algebra (1.2.1). Hence this algebra is symmetric.

One may conjecture that, more generally, $L^p(G, \omega)$ is a symmetric Banach $*$ -algebra, if (G, ω) satisfies (LPAIlg) and ω satisfies condition (S), or, equivalently, the GRS-condition. But this remains an open question.

Figure 1: The source's Theorem 3.9 and the conjecture immediately following it, source PDF page 12.

2 Proof intuition

Condition (S) is exactly strong enough to remove the weight from high convolution powers of a compactly supported function. If f is supported in U^m , then f^{*n} is supported in U^{mn} , so its weighted norm costs at most $s(mn)$. Young's inequality leaves one L^p factor and measures the other $n - 1$ factors in L^1 . Taking n th roots kills $s(mn)$ and recovers the L^1 spectral radius. Applying this to every polynomial in a self-adjoint f upgrades equality of radii to equality of spectra.

The general completion is delicate because compact truncations need not preserve spectral radius in a noncommutative Banach algebra. Two settings avoid this. Central elements of a discrete group possess *commuting* finite-support truncations. More decisively, if $G = H \times F$ with H abelian and F finite, the whole algebra is, up to an equivalent norm, a finite direct sum of matrix algebras over the already symmetric abelian weighted algebra. This proves the full conjecture for a genuinely nonabelian family.

3 The compact core is self-adjoint spectrally invariant

Write $r_{p,\omega}$ and r_1 for spectral radii in $L^p(G, \omega)$ and $L^1(G)$, using unitizations when needed.

Theorem 1 (Compact-core spectral invariance). *Assume (LPAlg) and condition (S). If $f \in L^p(G, \omega)$ has compact support, then*

$$r_{p,\omega}(f) = r_1(f).$$

If in addition $f = f^$, then*

$$\sigma_{L^p(G,\omega)}(f) = \sigma_{L^1(G)}(f) \subset \mathbb{R}.$$

In particular, the dense $$ -subalgebra $C_c(G)$ is Hermitian with the spectrum computed in the completed weighted algebra.*

Proof. Choose m with $\text{supp } f \subset U^m$. For $n \geq 2$,

$$\text{supp}(f^{*n}) \subset U^{mn}$$

and unweighted Young's inequality gives

$$\begin{aligned} \|f^{*n}\|_{p,\omega} &\leq s(mn) \|f^{*n}\|_p \\ &\leq s(mn) \|f^{*(n-1)}\|_1 \|f\|_p. \end{aligned}$$

Consequently,

$$r_{p,\omega}(f) \leq \lim_{n \rightarrow \infty} s(mn)^{1/n} \|f^{*(n-1)}\|_1^{1/n} \|f\|_p^{1/n} = r_1(f),$$

because $s(mn)^{1/n} = (s(mn)^{1/(mn)})^m \rightarrow 1$. Conversely, (LPAlg) and $\omega^{-q} \in L^1(G)$ give a continuous algebra embedding $L^p(G, \omega) \hookrightarrow L^1(G)$, hence $r_1(f) \leq r_{p,\omega}(f)$.

Now suppose $f = f^*$ and set $K = \sigma_{L^1(G)}(f)$. The group G has polynomial growth, so $L^1(G)$ is symmetric by Losert's theorem; hence $K \subset \mathbb{R}$. For every polynomial P with $P(0) = 0$, the element $P(f)$ remains compactly supported. The first part and spectral mapping give

$$r_{p,\omega}(P(f)) = r_1(P(f)) = \max_{z \in K} |P(z)|.$$

If $\lambda \in \sigma_{L^p(G,\omega)}(f)$, spectral mapping therefore yields

$$|P(\lambda)| \leq \max_{z \in K} |P(z)| \quad \text{for all such } P.$$

A compact subset of the real line is polynomially convex. In the nonunital case $0 \in K$, so polynomials vanishing at zero still separate every point outside K . Thus $\lambda \in K$. The reverse spectral inclusion follows from the algebra embedding, proving equality. $\square$

The theorem is stronger than a mere density statement: spectra are computed in the completed weighted algebra. It also isolates the remaining issue: spectral radius is not generally continuous under noncommuting norm approximation.

4 A completion upgrade for discrete central elements

Proposition 1. *Suppose in addition that G is discrete. If z is central in $L^p(G, \omega)$, then*

$$r_{p,\omega}(z) = r_1(z).$$

If $z = z^$, then its spectra in $L^p(G, \omega)$ and $L^1(G)$ coincide and are real.*

Proof. A central function on a discrete group is constant on conjugacy classes. Since $\omega \geq 1$, a weighted ℓ^p function must vanish on every infinite conjugacy class. It follows that z has finite-support central truncations z_N with $\|z - z_N\|_{p,\omega} \rightarrow 0$. Put $h_N = z - z_N$. All three elements commute. Spectral radius is subadditive on commuting elements, Theorem 1 applies to z_N , and the embedding into $\ell^1(G)$ has some norm C . Hence

$$\begin{aligned} r_{p,\omega}(z) &\leq r_{p,\omega}(z_N) + \|h_N\|_{p,\omega} \\ &= r_1(z_N) + \|h_N\|_{p,\omega} \\ &\leq r_1(z) + \|h_N\|_1 + \|h_N\|_{p,\omega} \\ &\leq r_1(z) + (C + 1)\|h_N\|_{p,\omega}. \end{aligned}$$

Letting $N \rightarrow \infty$ and using the reverse inequality proves equality. The same radius equality applies to every polynomial in a central z . When $z = z^*$, polynomial separation of its real L^1 spectrum, as in Theorem 1, gives equality of spectra. $\square$

5 A full nonabelian subcase

Theorem 2 (Finite direct factors). *Let H be a compactly generated locally compact abelian group of polynomial growth, let F be a finite group, and put $G = H \times F$. If (G, ω) satisfies (LPAlg) and condition (S), then*

$$L^p(G, \omega)$$

is a symmetric Banach $$ -algebra. In particular, this proves the source conjecture for nonabelian groups such as $\mathbb{R} \times S_3$, for every admissible GRS weight and not only for Pytlík-polynomial weights.*

Proof. Normalize Haar measure on the finite factor arbitrarily and define

$$\omega_H(h) = \omega(h, e_F), \quad B = L^p(H, \omega_H).$$

For $t \in F$, submultiplicativity gives

$$\frac{\omega_H(h)}{\omega(e_H, t^{-1})} \leq \omega(h, t) \leq \omega_H(h)\omega(e_H, t).$$

Because F is finite, these constants are uniform in t . Thus, as a Banach space with equivalent norms,

$$L^p(H \times F, \omega) = \bigoplus_{t \in F} B.$$

The restriction ω_H is symmetric and submultiplicative. In the convolution inequality for ω^{-q} on $H \times F$, retain only the $t = e_F$ summand. This gives

$$\omega_H^{-q} *_H \omega_H^{-q} \leq C \omega_H^{-q}$$

for a constant depending only on the Haar normalization of F . After the standard scalar renormalization, (H, ω_H) satisfies (LPAI). Condition (S) also restricts from $H \times F$ to H : use a generating set $V \times F$, where V generates H . The abelian theorem of [1] therefore shows that B is symmetric.

Convolution on the direct product identifies the Banach $*$ -algebra, up to the equivalent norm already noted, with the finite tensor product

$$B \otimes \mathbb{C}[F].$$

Finite-group Fourier theory gives a $*$ -algebra decomposition

$$\mathbb{C}[F] \cong \bigoplus_{\pi \in \widehat{F}} M_{d_\pi}(\mathbb{C}),$$

and hence

$$B \otimes \mathbb{C}[F] \cong \bigoplus_{\pi \in \widehat{F}} M_{d_\pi}(B).$$

Symmetry of Banach $*$ -algebras is preserved by finite matrix amplification and finite direct sums (the Leptin–Wichmann matrix theorem; see [3]). Therefore $L^p(H \times F, \omega)$ is symmetric. $\square$

6 Why the general completion remains open

Hulanicki’s transfer lemma requires equality of spectral radii for *every* element of $\mathcal{A}$ [2, Lemma 3.1]; an arbitrary dense core does not suffice, and spectral radius need not behave continuously under noncommuting compact truncations. The weighted- L^1 annular proof also requires a weighted first moment, whereas $L^p(G, \omega)$ supplies only a weighted p -moment and an unweighted first moment. Eight focused attempts confirmed this completion barrier. The general result is not claimed; a further upgrade needs a new all-elements inverse-closedness or representation theorem.

7 Verification and novelty status

No numerical evidence is used. The support estimate, Hulanicki hypothesis, finite-factor coordinates, and Haar-normalization constants were audited directly. On August 11, 2026, bounded index and web searches by exact title, quoted question, authors, weighted- L^p , GRS, and condition (S) found no later general solution or statement of Theorem 2. Novelty confidence is moderate pending specialist review, especially of the nonunital polynomial separation and finite matrix reduction.

References

- [1] Y. Kuznetsova and C. Molitor-Braun, *Harmonic analysis of weighted L^p -algebras*, *Expositiones Mathematicae* **30** (2012), 124–153; arXiv:1103.3610.
- [2] G. Fendler, K. Gröchenig, and M. Leinert, *Symmetry of weighted L^1 -algebras and the GRS-condition*, *Bull. London Math. Soc.* **38** (2006), 625–635.
- [3] T. W. Palmer, *Banach Algebras and the General Theory of $*$ -Algebras, Vol. II*, Cambridge University Press, 2001.